

PO9:

$$\text{Object size} = 1 \text{ Mbit/sec}$$

$$\text{Request rate} = 16 \text{ requests/sec}$$

$$\text{Internet delay} = 3 \text{ secs (RTT from institutional router to origin server)}$$

$$\text{Access delay} = \frac{\Delta \rightarrow \text{transmission delay}}{(1 - \Delta B) \rightarrow \text{request rate}}$$

$$\text{Total delay} = \text{Internet delay} + \text{access delay}$$

$$\begin{aligned} \text{Access delay} &= \frac{1000,000 / (15 \times 1000000)}{[(1000000 / (15 \times 1000000)) \times 16] - 1} \\ &= 1 \end{aligned}$$

(a)

$$\begin{aligned} \text{So, Total delay} &= 3 + 1 \\ &= 4 \text{ s} \end{aligned}$$

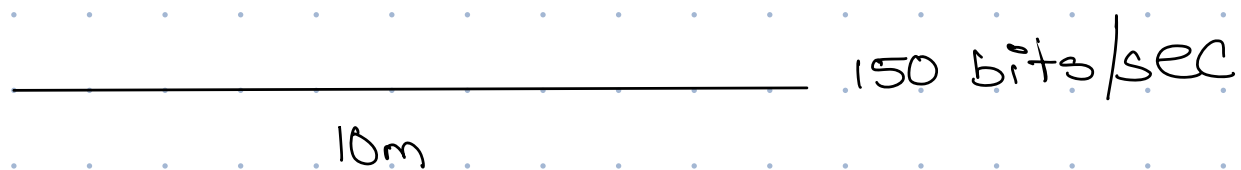
(b) As only 0.4 requests go on to the access link:

$$\begin{aligned} \text{Access delay} &= \frac{1000,000 / (15 \times 1000000)}{[(1000000 / (15 \times 1000000)) \times 16 \times 0.4] - 1} \\ &= 0.1162 \text{ s} \end{aligned}$$

$$\begin{aligned} \text{Total miss delay} &= 3 + 0.1162 \\ &= 3.1162 \text{ s} \end{aligned}$$

$$\begin{aligned} \text{So, Total delay} &= 0.6 \times 0 + 0.4 \times 3.1162 \\ &= 1.246 \text{ sec} \end{aligned}$$

P10:



data packet = 100000 bits
control packet = 200 bits

Total size = 100200 bits

Object size = 100 kbits

Total objects = 10

Parallel

$$T_p \text{ for data packets} = \frac{100000 \text{ bits}}{150 \text{ bits/sec}}$$
$$= 666.67 \text{ secs}$$

$$T_p \text{ for control packets} = \frac{200 \text{ bits}}{150 \text{ bits/sec}}$$
$$= 1.33 \text{ secs}$$

$$\text{Total delay} = 11 \times 666.67 + 11 \times 1.33 \text{ secs}$$
$$= 7,348.036 \text{ secs.}$$

Persistent:

$$\text{Total delay} = 11 \times 666.67 + 1.33$$
$$= 7334.7 \text{ secs}$$